

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	20 July 2026
Team ID	FinWise AI Team (SmartBridge SIP 2026 - SRM AP)
Project Name	FinWise AI - Intelligent Loan Eligibility, Credit Analysis & Financial Advisory Platform
Maximum Marks	4 Marks

Technical Architecture Overview

FinWise AI is a static, serverless web application. The browser directly calls the Groq AI API for financial-tip generation and a Google Apps Script Web App for data logging - there is no custom backend server or traditional database.

Table 1: Components & Technologies

S.No	Component	Description	Technology
1.	User Interface	How the user interacts with the application - dashboard, forms, results	HTML5, CSS3 (glassmorphism), Vanilla JavaScript
2.	Application Logic-1	Loan eligibility rule evaluation & eligible amount calculation	JavaScript (app.js / eligibility.js)
3.	Application Logic-2	Credit score classification (Excellent/Good/Poor)	JavaScript (credit-score.js)
4.	Application Logic-3	EMI computation using reducing-balance formula	JavaScript (emi.js)
5.	AI Recommendation Engine	Generates personalised financial guidance from user inputs	Groq AI Chat Completions API (LLaMA-based models, free tier)
6.	Data Storage	Stores each user submission as a row for tracking/analytics	Google Sheets
7.	Serverless Backend	Receives POST requests from the browser and writes to Sheets	Google Apps Script (Web App / doPost)
8.	External API	Purpose: real-time AI-generated financial advice	Groq Cloud API
9.	Infrastructure / Hosting	Static site hosting and deployment	GitHub Pages
10.	Version Control	Source control and collaboration	Git & GitHub

Table 2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No paid frameworks used; front-end built with vanilla web technologies	HTML5, CSS3, JavaScript (ES6)
2.	Security Implementation	HTTPS delivery via GitHub Pages; client-side input validation	GitHub Pages TLS, JS validation

S.No	Characteristics	Description	Technology
3.	Scalable Architecture	Stateless static front-end; new financial modules can be added as new pages	Modular multi-page structure (index/eligibility/credit-score/emi/tips)
4.	Availability	24x7 static hosting with no server to maintain	GitHub Pages CDN
5.	Performance	Rule-based logic runs instantly client-side; AI responses returned within seconds	Groq's low-latency LPU inference